Basic Principles of Medicine 1
Module: Foundations to Medicine
Lecture No: 14

Lecture Title:

Translation and Post-translational Modification

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Objectives: DLA: Genetic code and Translation

- Before lecture ‘Translation and Post-translational Modification’

SOM.MK.I.BPM1.1.FTM.3.BCHM.0064	Explain why the genetic code is collinear with DNA, triplet in nature, redundant (degenerate), and non-overlapping.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0065	Define the terms: reading frame and frame-shift mutation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0066	Interpret the standard genetic code table. Identify the initiation codon and the three termination codons.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0067	Explain the universal nature of the genetic code (with minor differences in mitochondria) and predict the sequence of a peptide if the coding strand of the DNA is provided.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0068	Discuss how protein synthesis proceeds from the N-terminal to their C-terminal ends and explain how this creates a naming convention for peptides.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0050	List and describe the major function of the following types of RNA in cells: mRNA, tRNA, rRNA, snRNA, miRNA.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0070	Describe the subunit composition prokaryotic (70S) and eukaryotic (80S) ribosomes. Recognize the roles of A, P, and E sites on the ribosome.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0071	Describe the structure, function, and charging of tRNAs.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0072	Describe the codon/anticodon interaction and discuss the wobble hypothesis.

Translation and post-translational modifications

Lecture Objectives

SOM.MK.I.BPM1.1.FTM.3.BCHM.0073	Describe the sequence of events that occurs during translation in prokaryotes (initiation, elongation and termination) and list the major differences between prokaryotic and eukaryotic translation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0074	Explain how diphtheria toxin interferes with eukaryotic translation. Explain the role of miRNA as inhibitors of translation.
SOM.MK.I.BPM1.1.FTM.3.BCHM.0069	Explain the mode of action of common antibiotics that interfere with translation: Initiation inhibitors (streptomycin), Elongation inhibitors (Tetracycline, chloramphenicol, erythromycin, puromycin, cycloheximide).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0075	Review examples of post-translational modifications: Zymogen activation (trypsinogen to trypsin), Serine/threonine phosphorylation (regulation of enzymes in metabolism), Tyrosine phosphorylation (Insulin receptor), O-linked glycosylation, N-linked glycosylation and Lipid anchoring (farnesyl groups to Ras).
SOM.MK.I.BPM1.1.FTM.3.BCHM.0076	Describe proteolytic processing and post-translational modifications using insulin as an example.

Pre-lecture Reading

View DLA videos:
Genetic Code and Translation

Recommended Reading

- **Questions:** 32.1-32.3, 32.5-32.9

- Lippincott's Biochemistry 9th Edition: Chapter 32: Protein Synthesis https://premiumbasicsciences.lwwhealthlibrary.com/content.aspx?sectionid=260214699&bookid=3402

- **Lippincott's Biochemistry**: Chapter 32 Protein Synthesis | Lippincott Illustrated Reviews: Biochemistry, 9e | Premium Basic Sciences | Health Library

Additional Resources

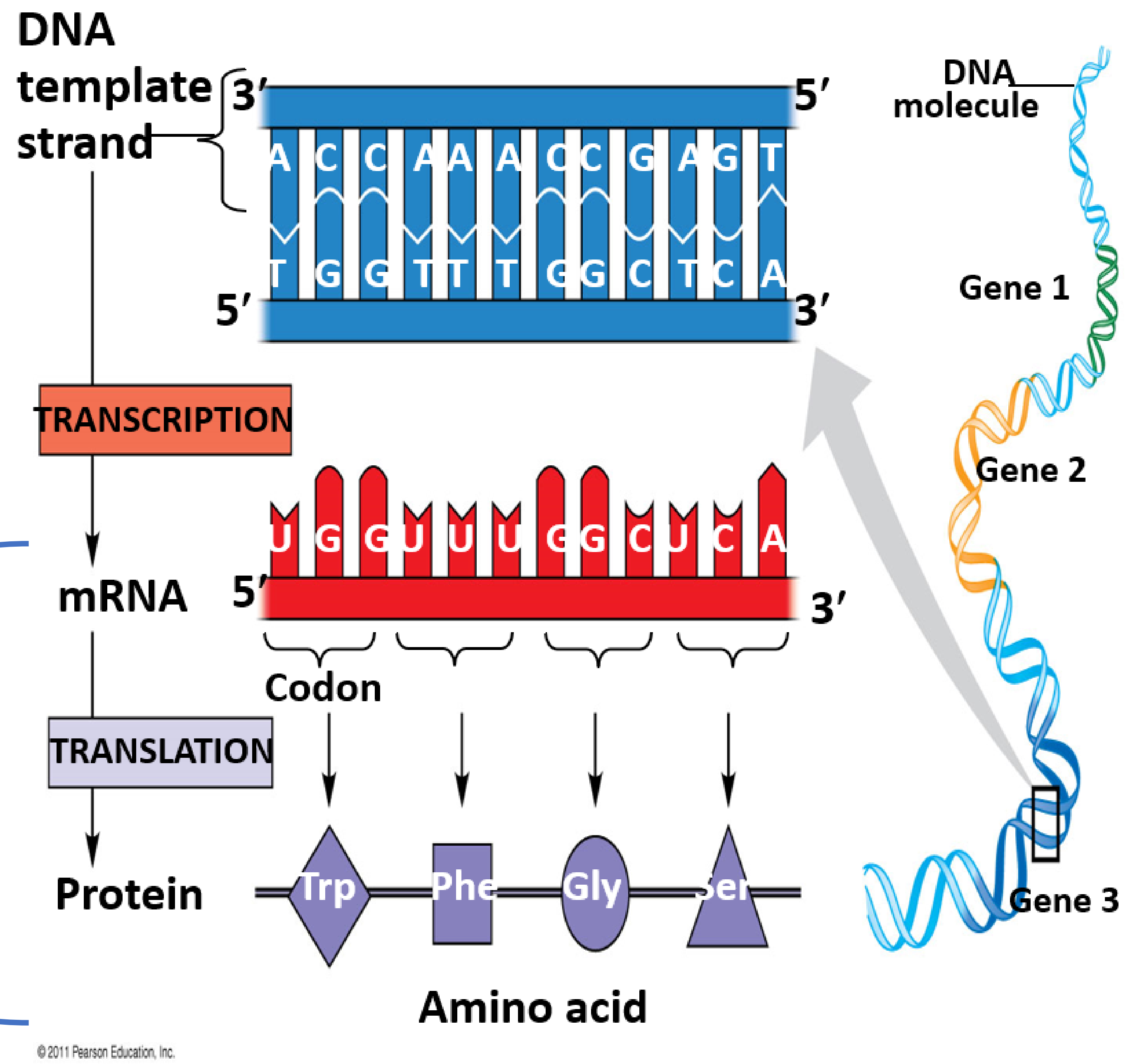
YouTube videos:

- The big picture of translation:
  - http://www.youtube.com/watch?v=5bLEDd-PSTQ
    - 3 minutes 32 seconds.
- Interpreting the genetic code:
  - https://www.youtube.com/watch?v=-Ht81lHiJac
    - 4 minutes 16 seconds.

Translation

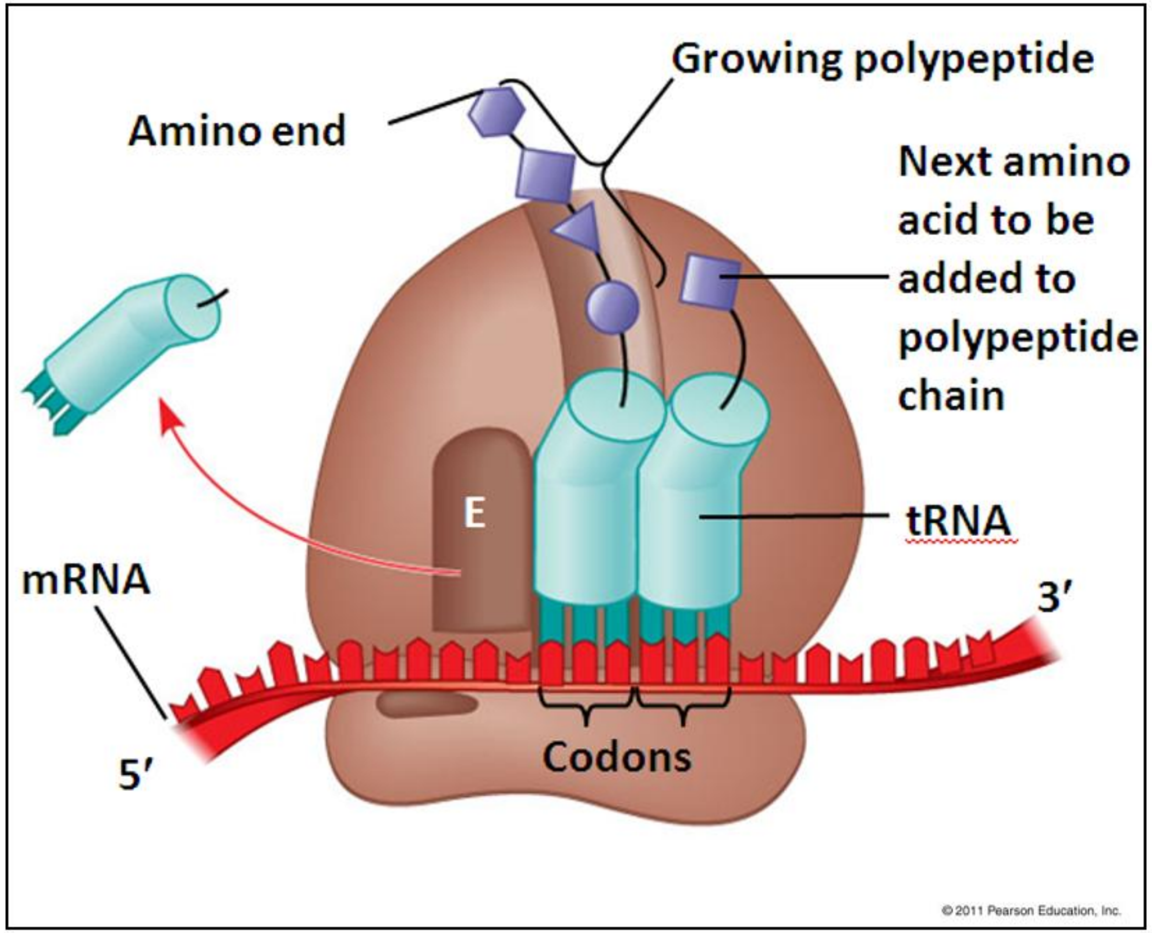
Formation of a polypeptide polymer from an RNA template

- ➢ mRNA
- ➢ Translated to
- ➢ Polypeptide



Translation: Requirements

- The biological polymerization of amino acids into a polypeptide chain (Translation) requires:
  - Messenger RNA (mRNA)
  - Ribosomes
  - Charged transfer RNA (tRNA)
  - Initiation factors
  - GTP



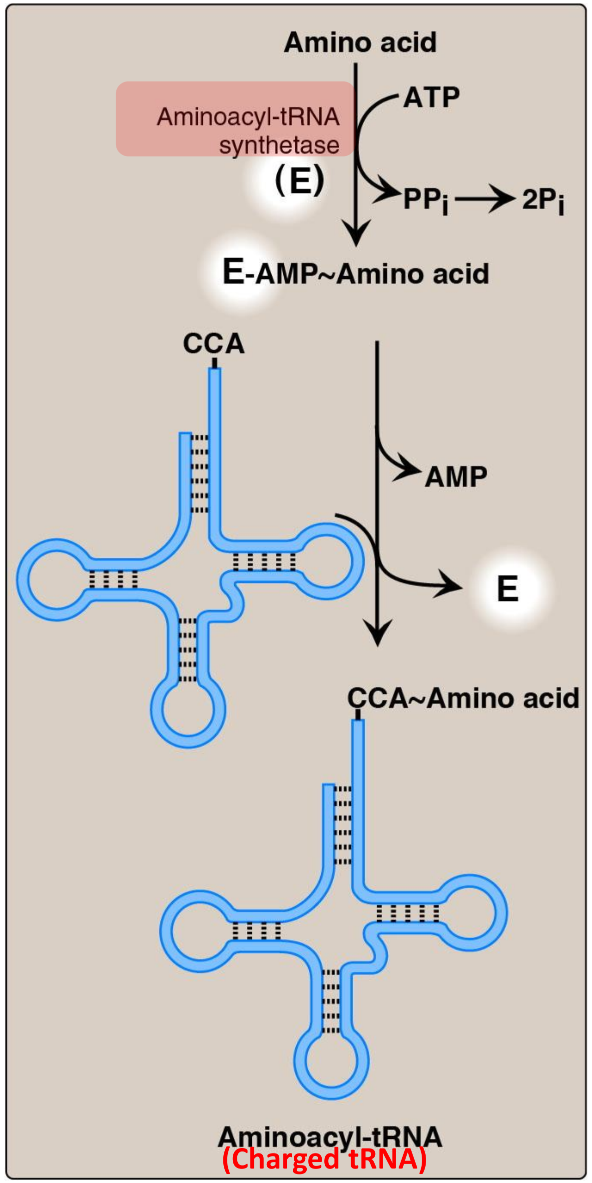
Schematic Model of Ribosome with mRNA, tRNA and growing Polypeptide chain

Translation: The Process

1. Activation of the monomer
2. Initiation
3. Elongation
4. Termination
5. Processing the polymer

Translation: the RNA-directed synthesis of a polypeptide

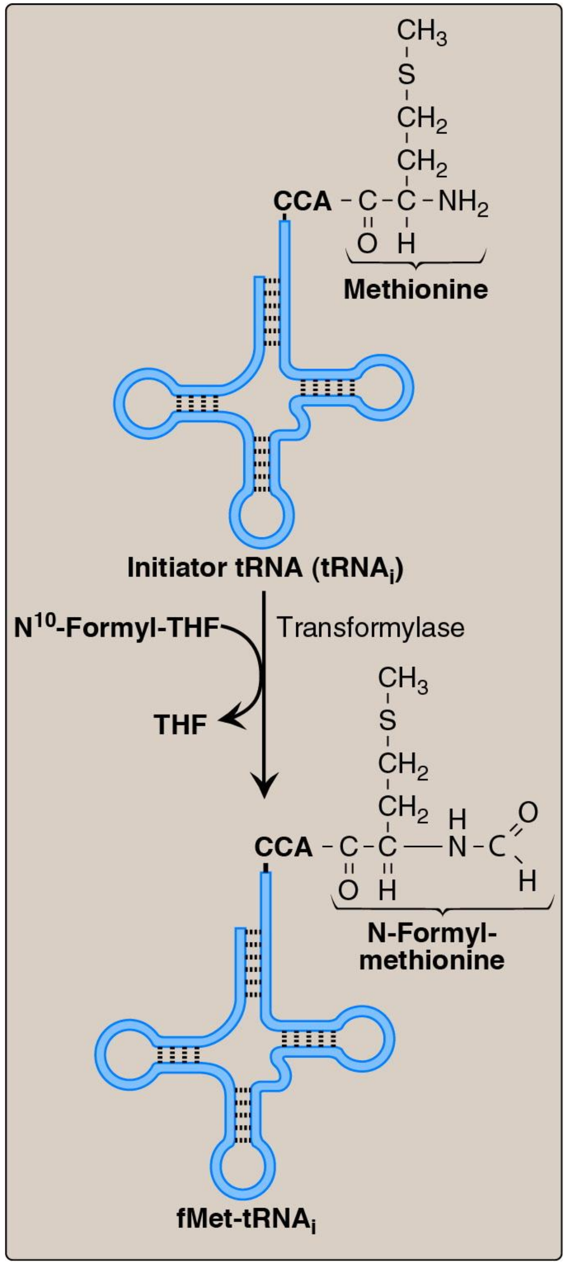
We will describe translation in prokaryotes and provide notes regarding the eukaryotic system when appropriate.



Amino acids activated by attachment to tRNA

- **Charging the tRNA** is a two step process:
    1. Enzyme bound amino-acid-adenylate
    2. Formation of the aminoacyl-tRNA
- Reaction is driven by hydrolysis of pyrophosphate
- tRNA to which amino acid is attached is called the "charged tRNA"

Generation of the initiator N-formylmethionyl-transfer RNA (fMet-tRNA$_i$).



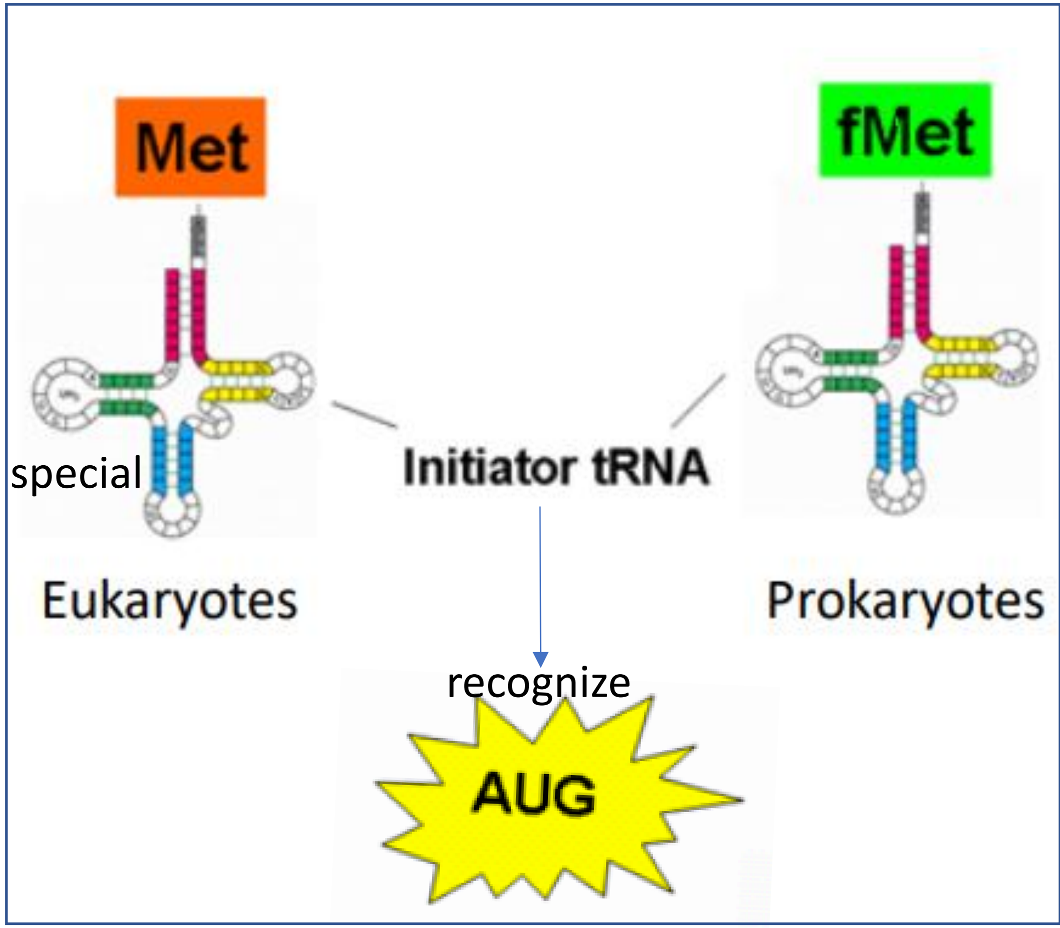
- N-formylmethionine (fMet) as the first amino acid – **in prokaryotes & in mitochondria.**
- This special tRNA$_i$ (fMet-tRNA$_i$) is recognized differently by the ribosome – **allows initiation**
- Prokaryotes have 2 tRNAs for methionine:
  - one allows formation of fMet,
  - the other recognizes internal AUG codons.

Use of Special Met-tRNA$_i$ as Initiator tRNA

The idea is to get the *first* MET containing tRNA$_i$ into the P-site to allow initiation

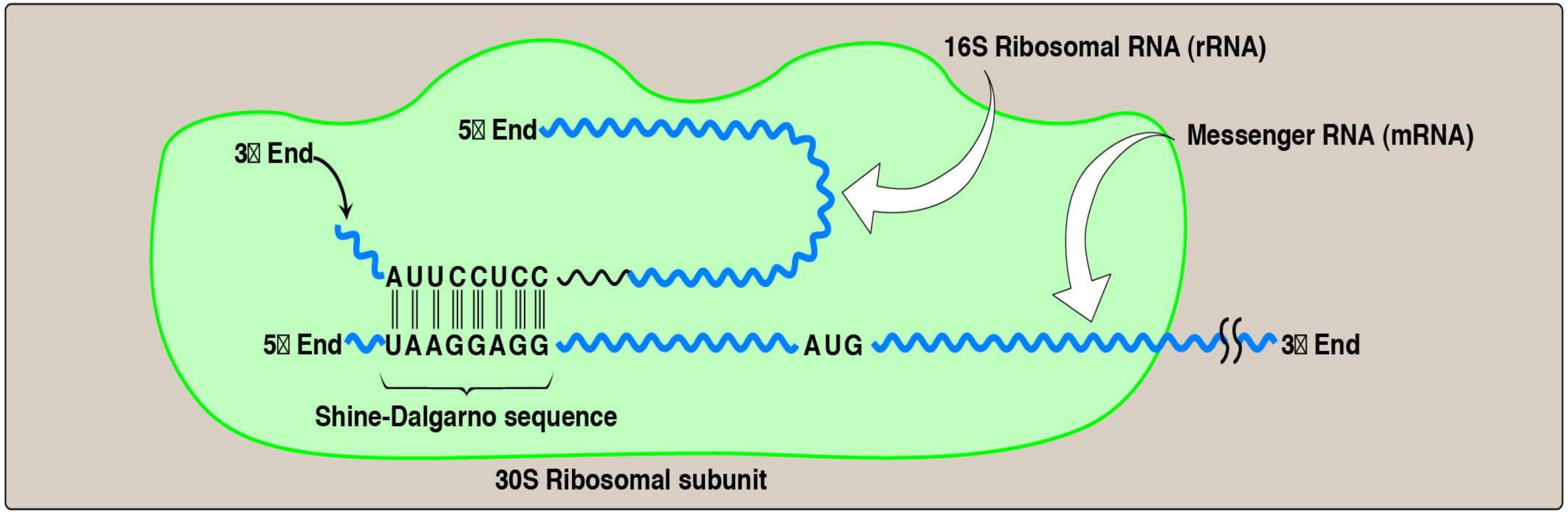
- **Prokaryotes & Mitochondria of Eukaryotes:**
  - Two tRNAs that recognize AUG
    1. Formylated MET for first codon
    2. Normal Met-tRNA for internal codons

- **Eukaryotes:**
  - Two tRNAs that recognize AUG
    1. The first codon also uses MET, and it has a special tRNA for this first codon (but the MET amino acid is not formylated).
    2. Normal Met-tRNA for internal codons



Initiator tRNA always carries methionine

Correct Alignment of AUG codon with respect to the ribosome



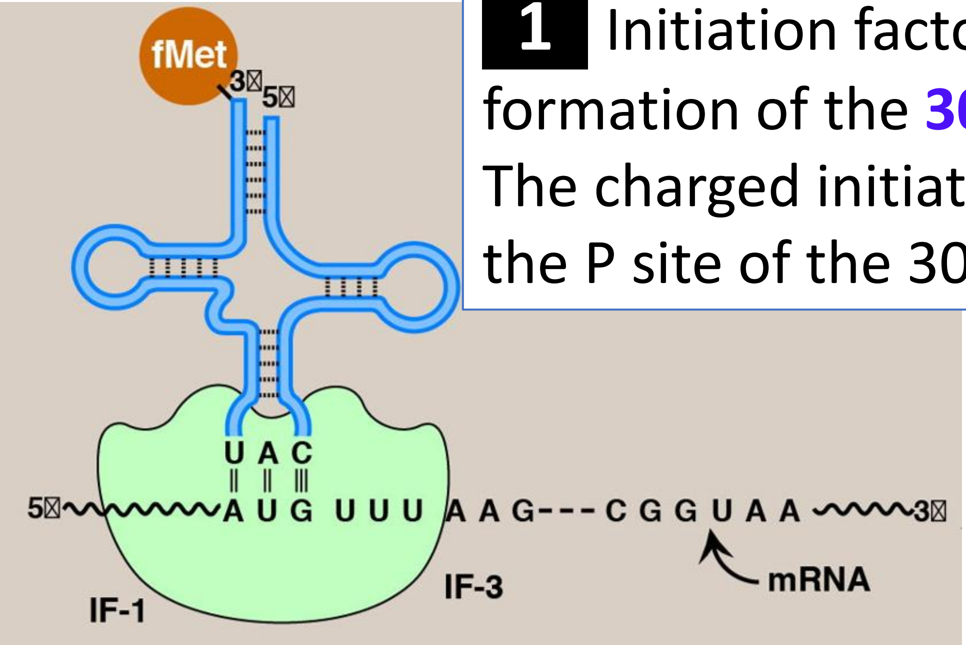
Complementary binding between prokaryotic mRNA Shine–Dalgarno sequence and 16S rRNA.

- **Prokaryotes:** Shine Dalgarno sequence is purine rich and resides a few (5-10) bases 5’ to the start codon
- **Eukaryotes:** lack Shine Dalgarno sequence, therefore eukaryotic small ribosome binds close to **the cap at the 5’ end**, **scans** until it encounters the AUG start codon.

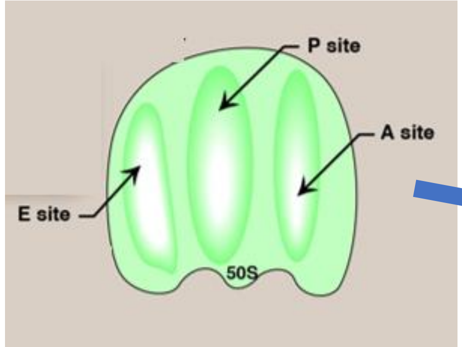
Protein Synthesis

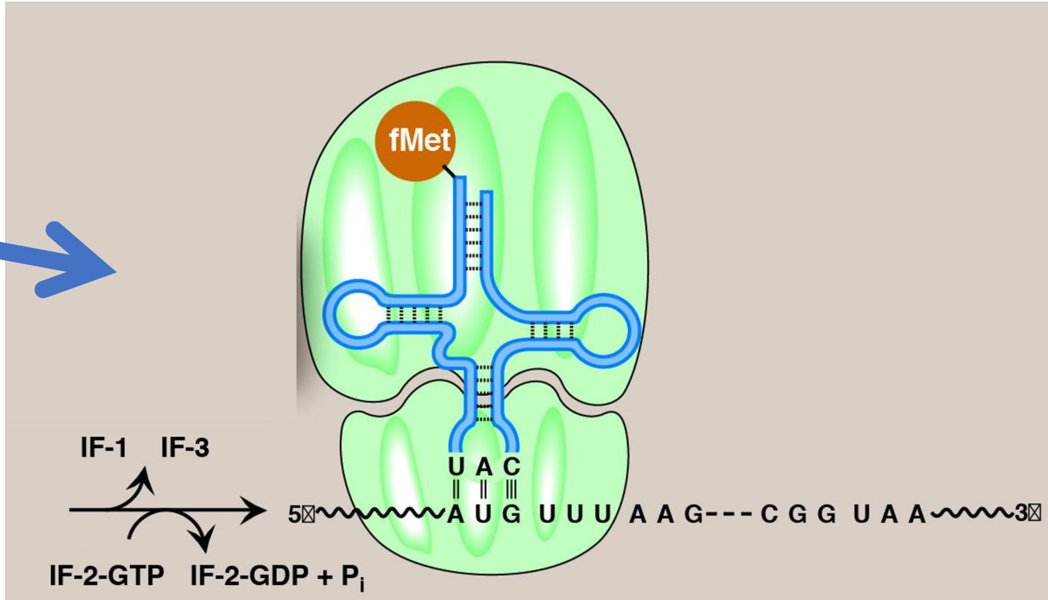
INITIATION

1 Initiation factors (IFs) aid in the formation of the **30S initiation complex**. The charged initiator tRNA is brought to the P site of the 30S subunit by IF-2-GTP.



2 GTP on IF-2 is hydrolyzed and initiation factors are released when the 50S subunit arrives to form the **70S initiation complex**.



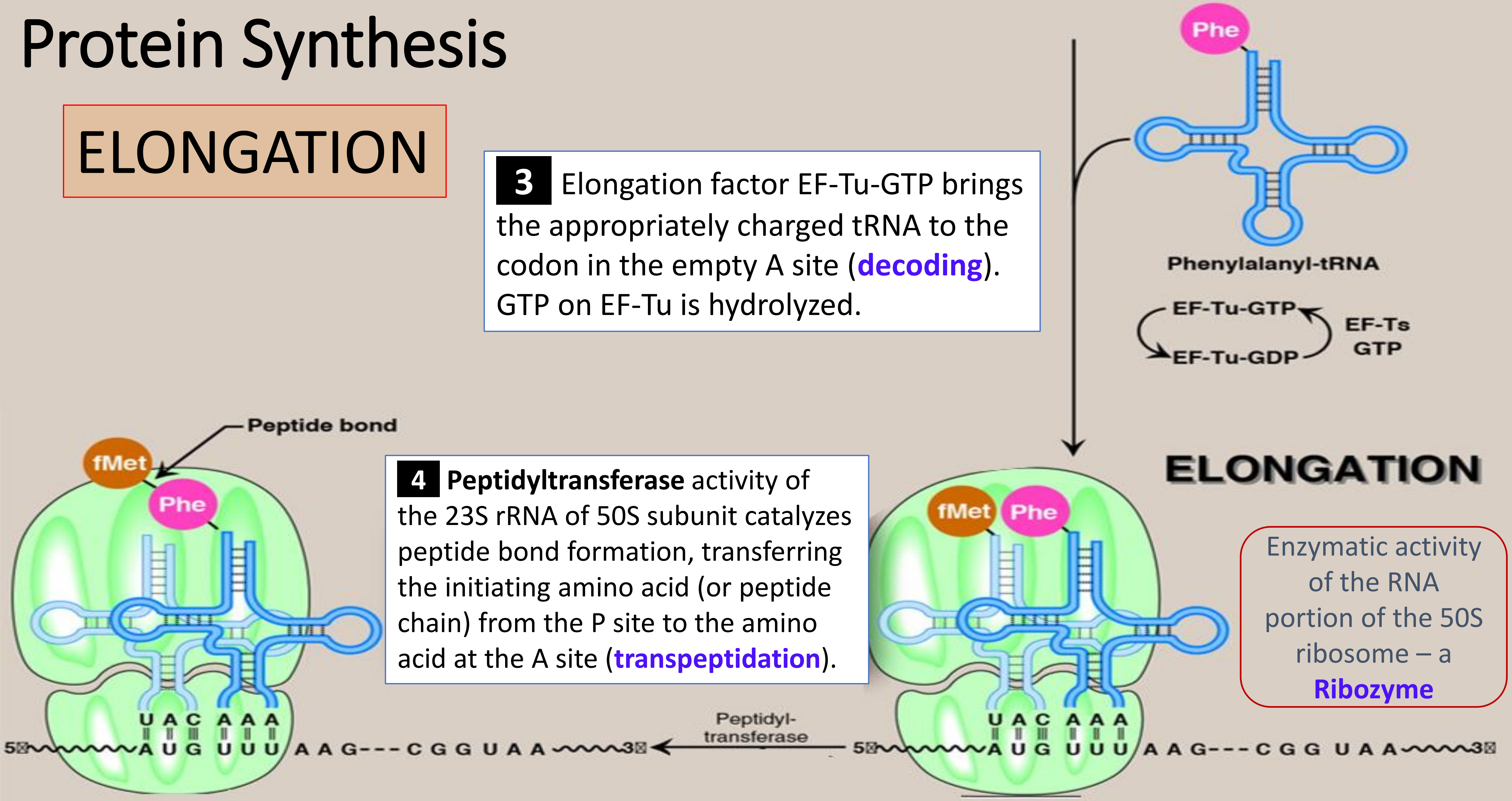


Protein Synthesis

ELONGATION

3 Elongation factor EF-Tu-GTP brings the appropriately charged tRNA to the codon in the empty A site (**decoding**). GTP on EF-Tu is hydrolyzed.

4 **Peptidyltransferase** activity of the 23S rRNA of 50S subunit catalyzes peptide bond formation, transferring the initiating amino acid (or peptide chain) from the P site to the amino acid at the A site (**transpeptidation**).

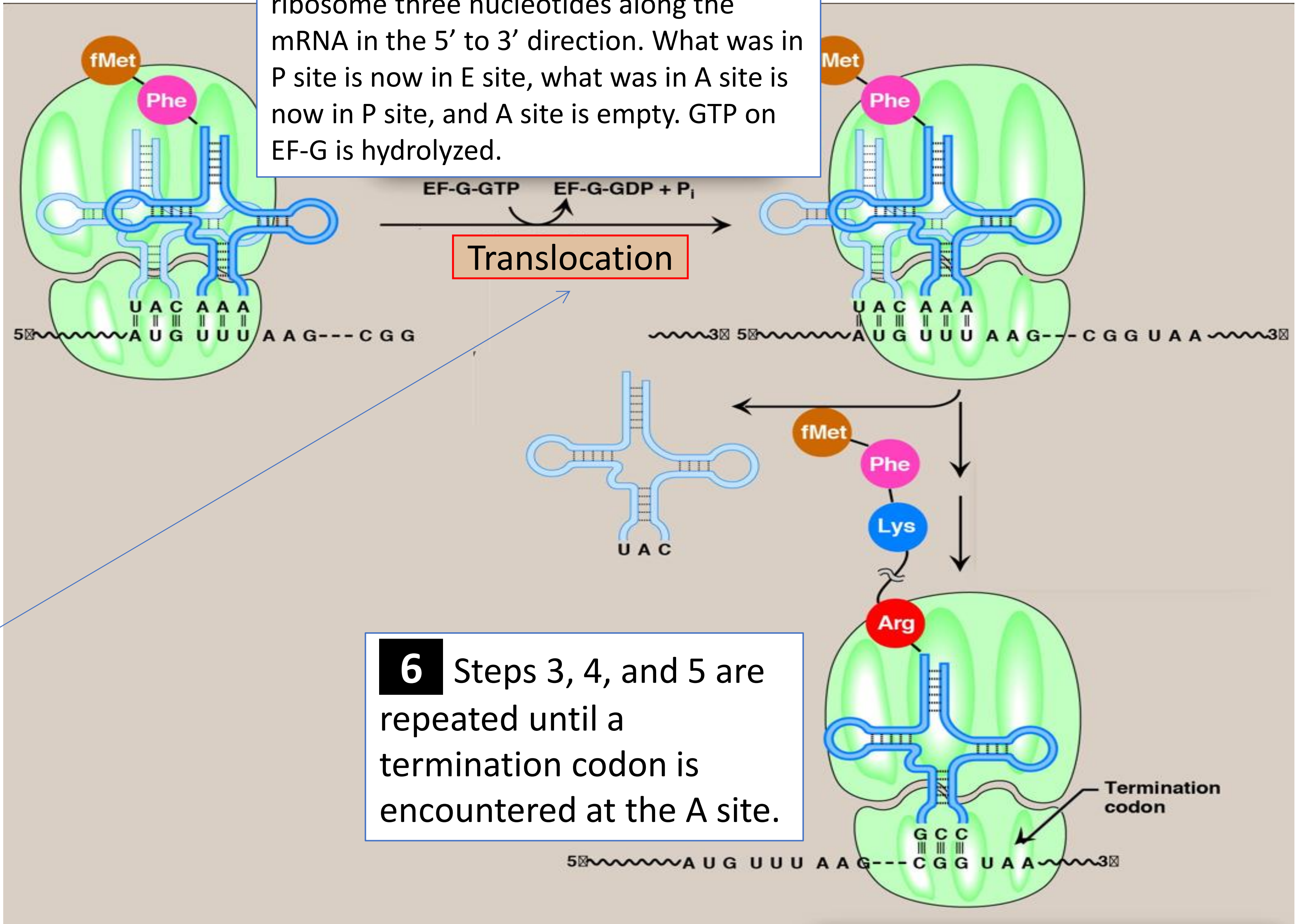


Enzymatic activity of the RNA portion of the 50S ribosome – a **Ribozyme**

Protein Synthesis

5 EF-G-GTP facilitates movement of the ribosome three nucleotides along the mRNA in the 5' to 3' direction. What was in P site is now in E site, what was in A site is now in P site, and A site is empty. GTP on EF-G is hydrolyzed.

TRANSLOCATION



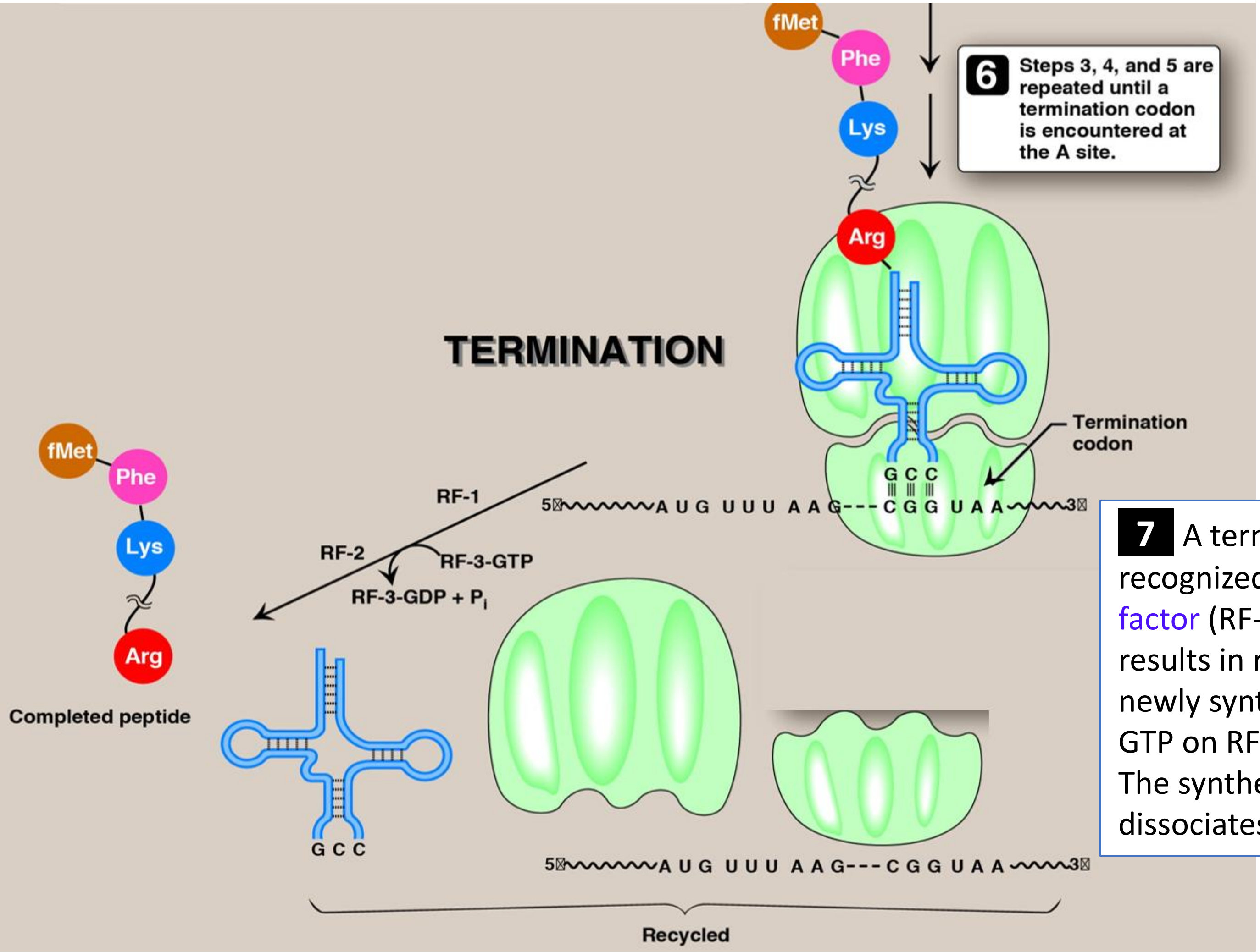
- EF-G is the prokaryotic protein
- In eukaryotes, EF-G is called EF-2

6 Steps 3, 4, and 5 are repeated until a termination codon is encountered at the A site.

Protein Synthesis

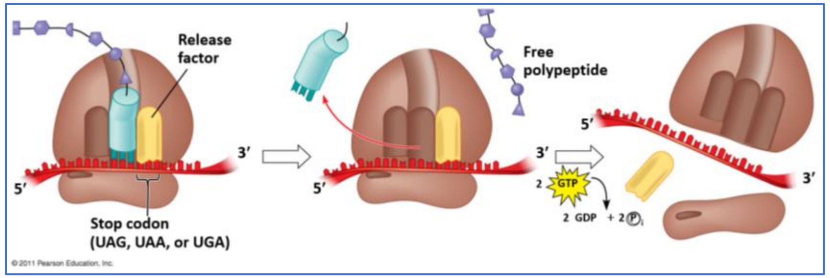
TERMINATION

- **Stop codon has no tRNA**
- Stop codon causes elongation to stall
- **Peptide is held in the P-site attached to the tRNA**
- **Release Factor** (RF) diffuses into the A-site
- RF allows **peptidyl transferase** to **cleave ester bond between tRNA and the peptide**

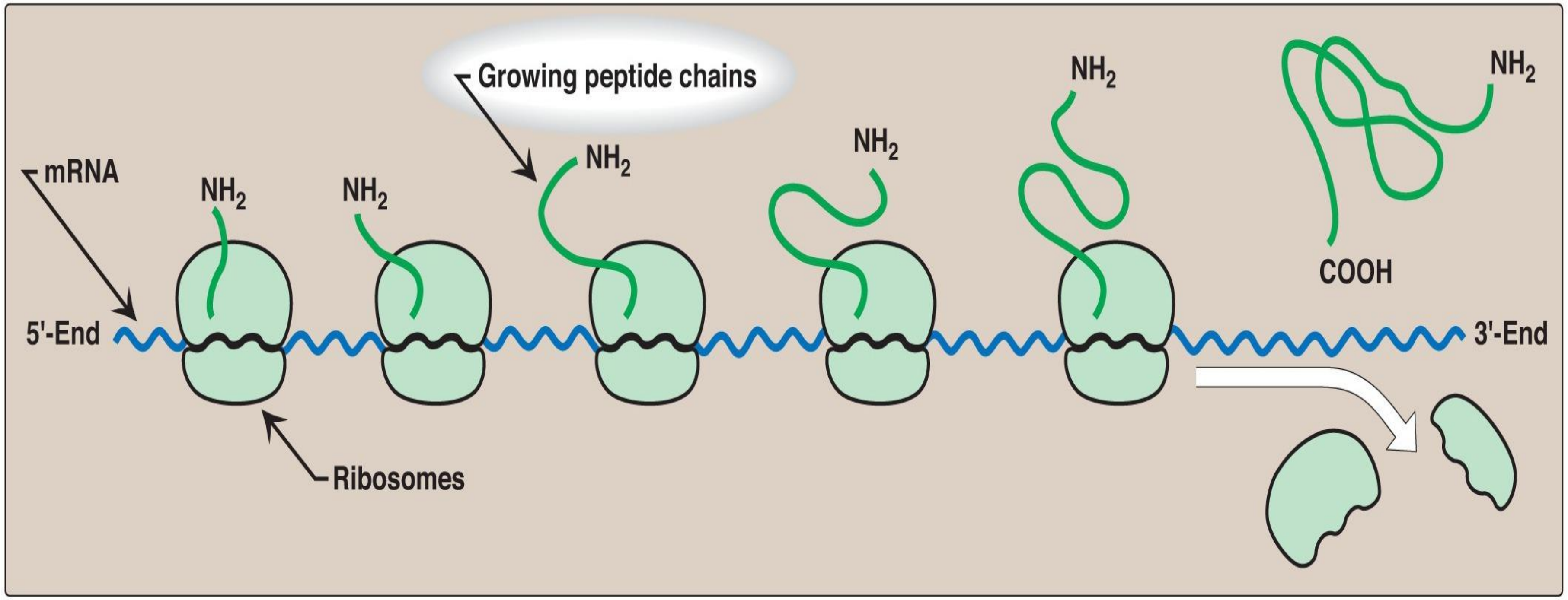


7 A termination codon is recognized by a release factor (RF-1 or RF-2), which results in release of the newly synthesized protein. GTP on RF-3 is hydrolyzed. The synthesizing complex dissociates.

- After **release of the peptide**, the **ribosome dissociates** into its subunits which may then begin a new round of translation.

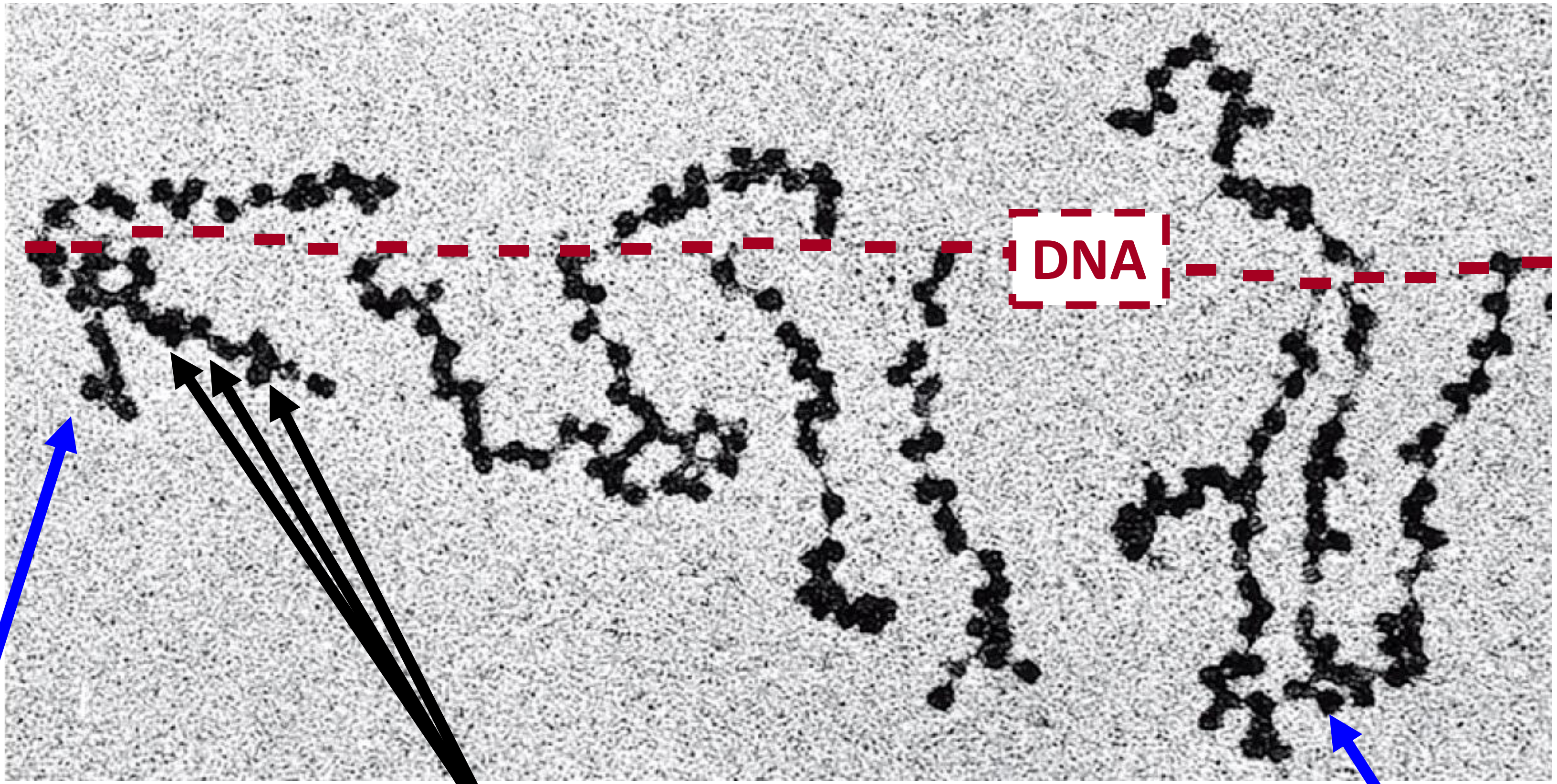


Polyribosome



- Ribosomes simultaneously translating one mRNA
- A number of ribosomes can translate a single mRNA simultaneously, forming a **polyribosome** (or **polysome**)
- Polyribosomes enable a cell to make many copies of a polypeptide very quickly.

Transcription and Translation is Coupled in Prokaryotes



Electron microscopy showing coupled transcription and translation in prokaryotes

Short transcripts as RNA polymerase begins transcription

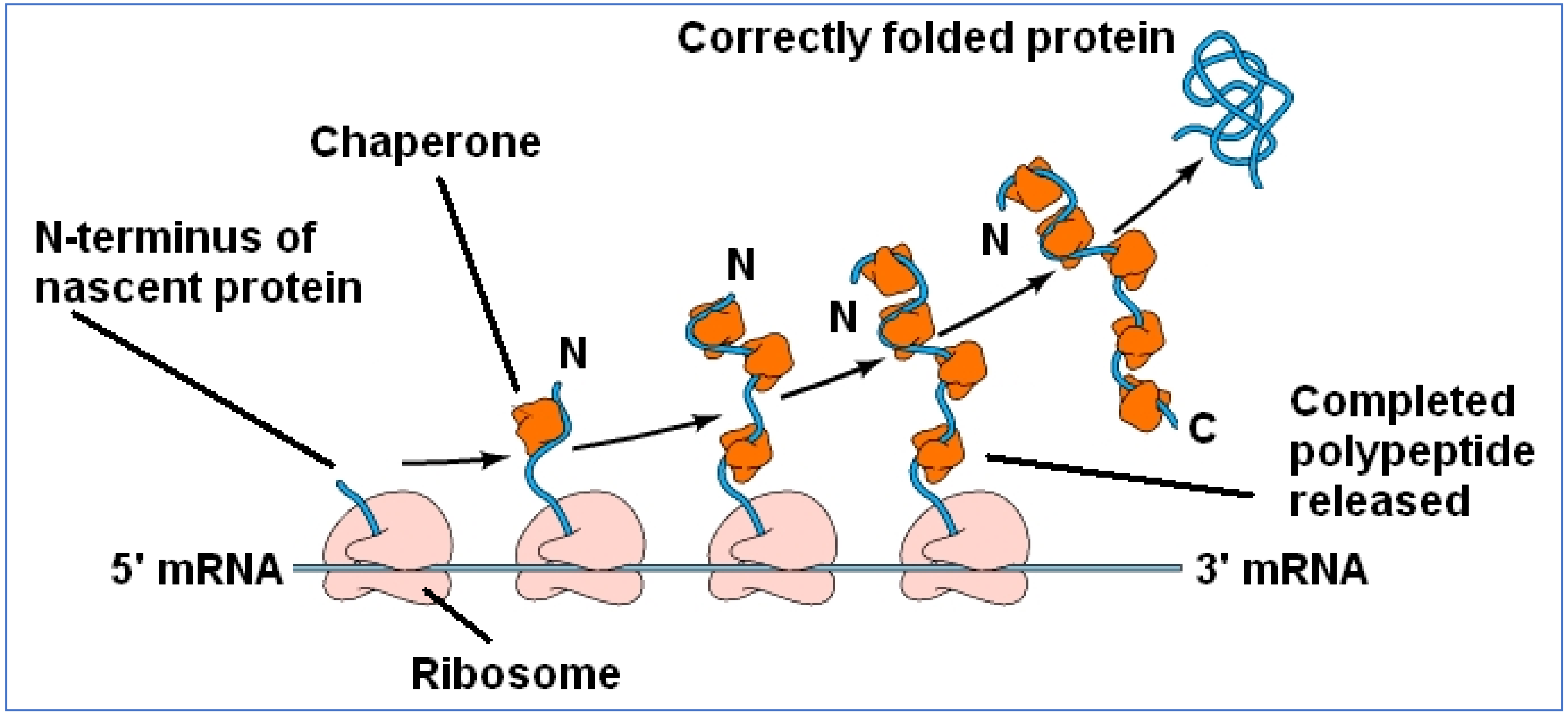
Polyribosome formation on the RNA transcript

longer transcripts as RNA polymerase nears the end of the gene

Proteins are not labeled, so we don't see them

Protein Folding

- Protein folding is the process whereby proteins acquire their mature functional (native) structure.
- Often begins co-translationally
- Occurs spontaneously or facilitated by chaperones.



Chaperones ensure that only a limited number of folds are available to a newly synthesized protein.

Some major differences between Prokaryotic & Eukaryotic Translation

Feature	**Prokaryotes**	**Eukaryotes**
Initiator tRNA	Formylated	Special, but not formylated
RNA	Polycistronic	Monocistronic
Translation start site	May select an internal AUG	Typically start translation at first AUG site
Transcription and Translation	Coupled	Not coupled because of nuclear membrane

Typically, methionine is NOT found as the first amino acid on a mature protein – post translationally modified

Compounds Affecting Protein Synthesis

- **Diphtheria toxin** inactivation of EF-2 by ADP-ribosylation

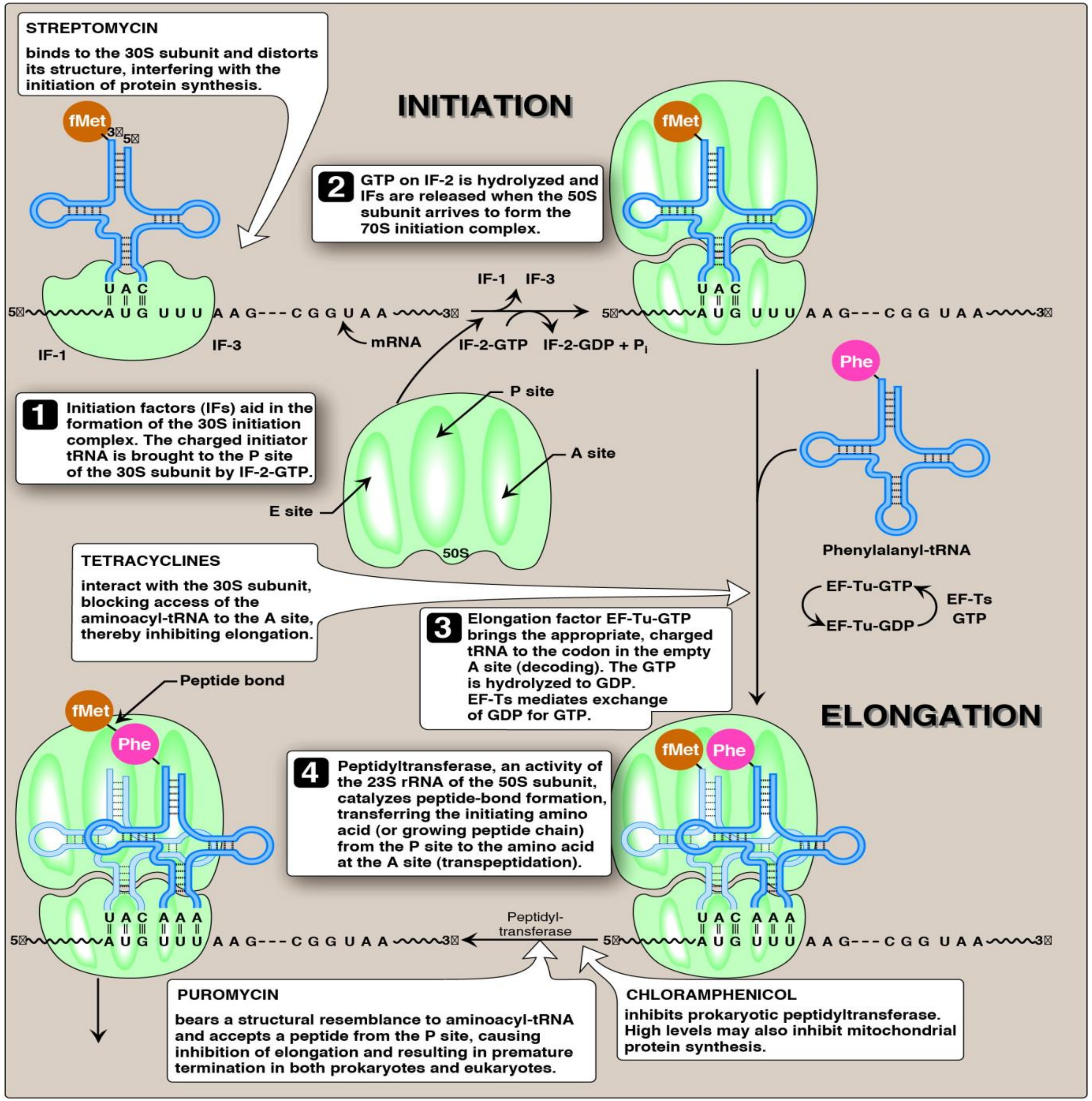
Characteristic membranous pharyngitis

Toxin A is produced by a lysogenic bacteriophage that infects *Corynebacterium diphtheriae*. The toxin Catalyzes the transfer of ADP-ribose to host cells EF-2, inactivating it (prevents translocation) & inhibiting protein synthesis.

- **<u>Antibiotic Inhibition of initiation</u>**
  - **Streptomycin** (aminoglycoside)
    - Prevents assembly of ribosome (binds to 30s subunit)
- **<u>Antibiotic Inhibition of elongation</u>**
  - **Tetracycline** - four (tetra) ring (cyclic) structure
    - Block elongation by preventing aminoacyl-tRNA access to the A-site
  - **Erythromycin** (macrolide)
    - Binds to the 50S subunit of the complete (70S) ribosome
    - Blocks ribosome translocation
  - **Chloramphenicol**
    - Inhibits peptidyl transferase activity in prokaryotes
    - At high levels, may inhibit mitochondrial translation
  - **Cycloheximide**
    - Inhibits eukaryotic peptidyl transferase activity
  - **Puromycin**
    - Causes premature termination of translation in both prokaryotes and eukaryotes

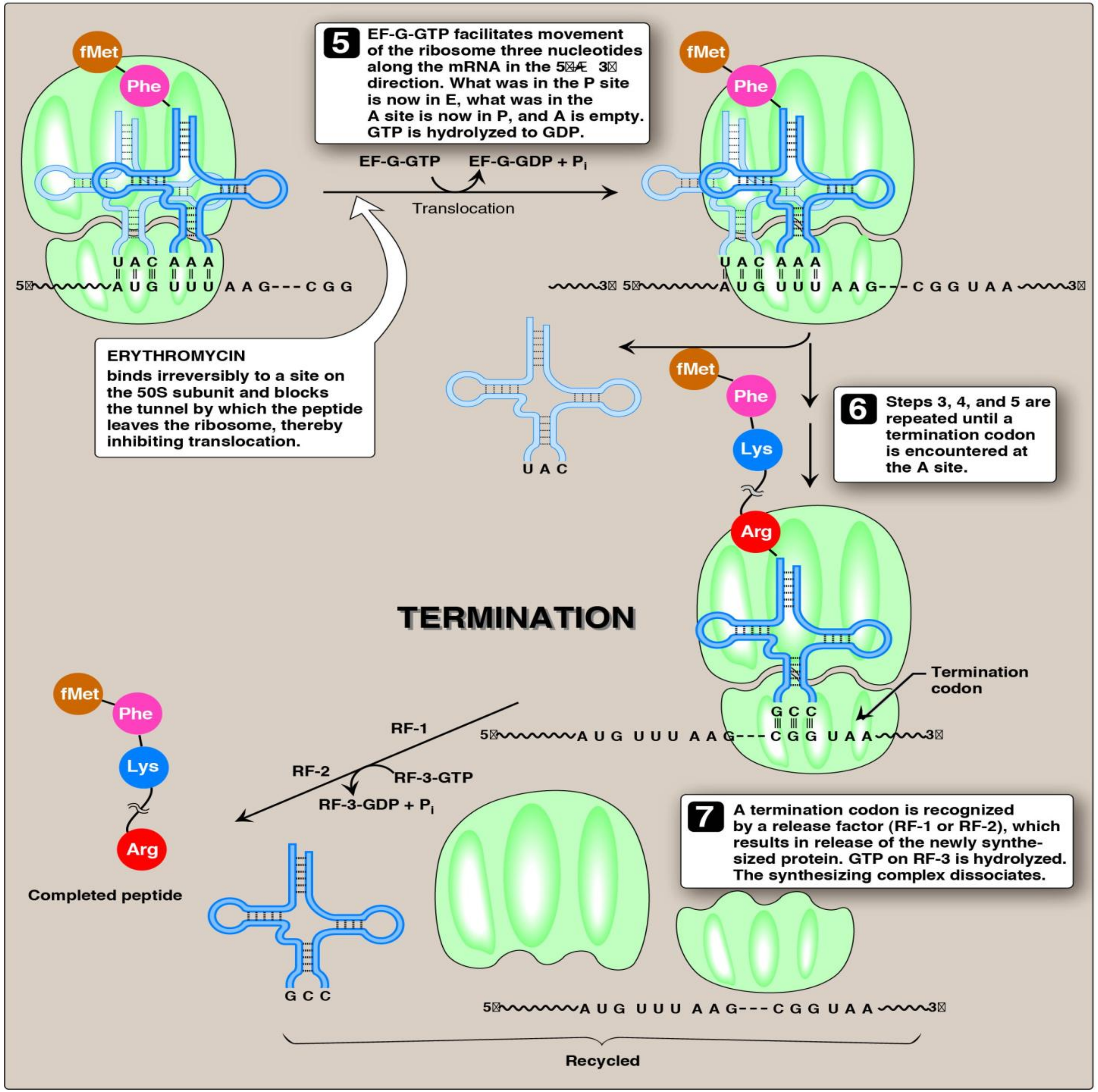
There are many more

Inhibitors of Protein Synthesis



Inhibitors of Protein Synthesis
-continued

- EF-G is the prokaryotic protein
- In eukaryotes, EF-G is called **EF-2 (inactivated by diphtheria toxin)**



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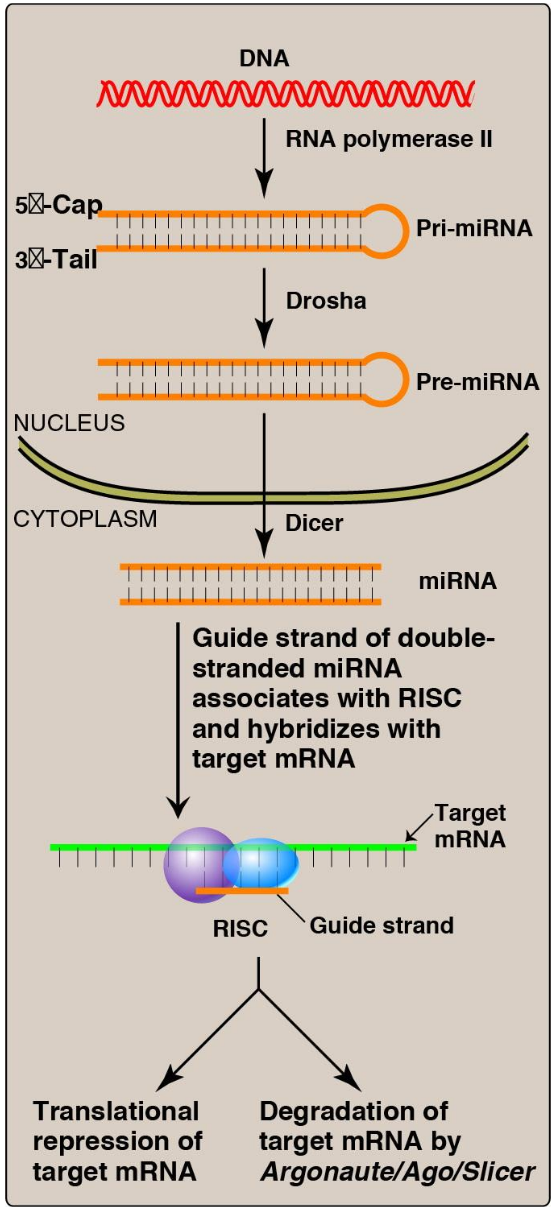
RNA Interference (RNAi) & Inhibition of Translation

- **miRNA as Inhibitors of Translation:** The extent of complementarity between the target mRNA and the miRNA determines the final outcome:
  - Perfect complementarity results in **mRNA degradation**.
  - Important role in processes like cell proliferation, differentiation, and programmed cell death (apoptosis).
- **Treatment of hereditary transthyretin-mediated amyloidosis (hATTR) by RNAi:** In 2018, the first RNAi-based therapy was approved to treat peripheral nerve disease (polyneuropathy) in patients with hATTR due to a mutation in the gene encoding transthyretin (TTR).
  - The siRNA-based drug, patisiran, prevents the production of abnormal TTR protein and reduces the buildup of amyloid deposits containing TTR that form in peripheral nerves and in the heart.



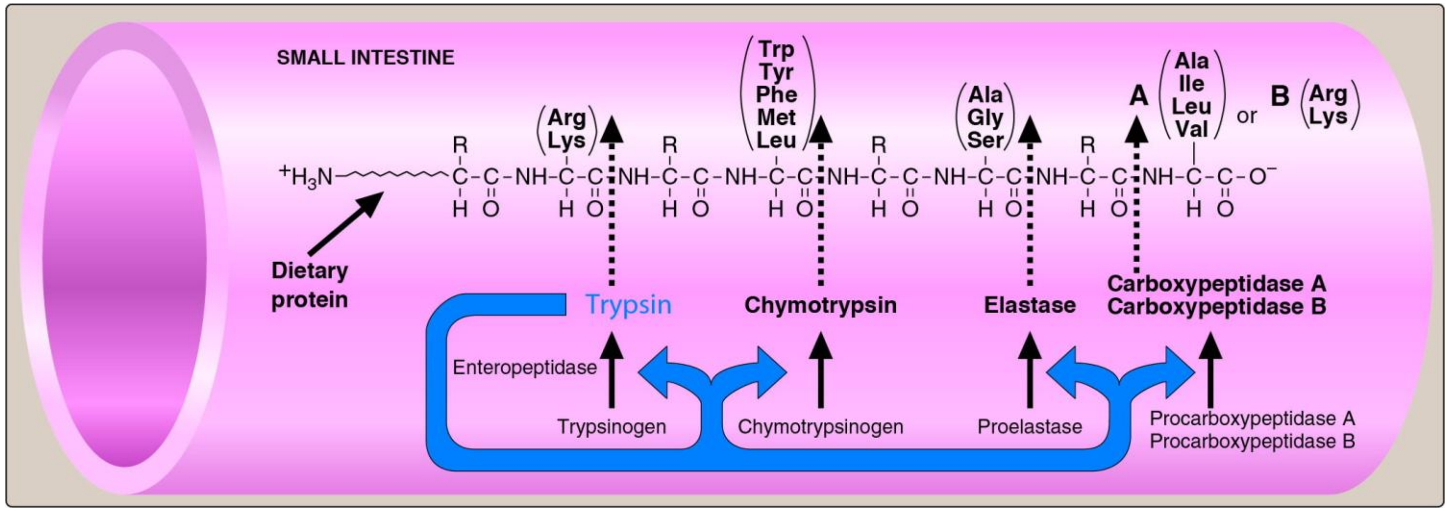
Production and action of microRNAs (miRNA)
Pri = primary; RISC = RNA-induced silencing complex.
Lippincott Biochemistry p525

Post-translational Modifications of Proteins

- Some proteins are released from the ribosome nearly ready to function, while others undergo a variety of post-translational modifications.
- Post-translational modification may occur as the polypeptide is being translated or after translation is completed and may be reversible or irreversible.
- Post-translational modification of a protein may result in its conversion to a functional form, its direction to a specific sub-cellular compartment, its secretion from the cell, alteration of its activity or stability.
- **Major types of post-translational modifications**:
    1) **Protein folding** (e.g. formation of disulphide bridges).
    2) **Covalent alterations** (e.g. phosphorylation, glycosylation, hydroxylation).
    3) **Proteolytic processing** (trimming: e.g. zymogen activation).
    4) **Addition of prosthetic groups** (e.g. Heme of hemoglobin).
    5) **Prenylation** (e.g. lipid anchoring, farnesylation).
    6) **Protein degradation** (e.g. ubiquitination and degradation by the proteasome complex)
- We will discuss a few!

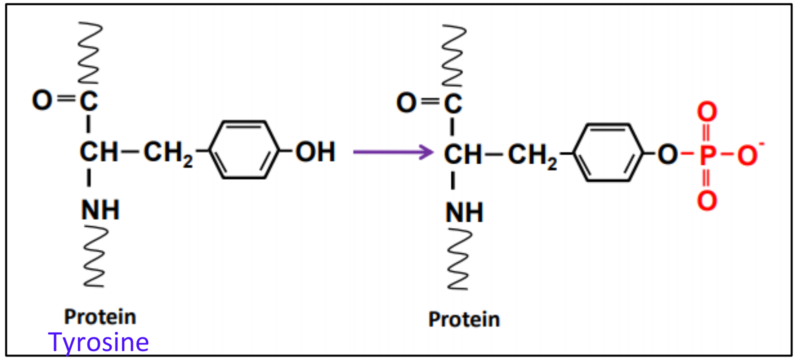
Zymogen Activation

- **Zymogen** (or proenzyme): an inactive enzyme precursor
  - Activated within an organism into active enzymes
  - Activation by enzymatic cleavage of peptide bonds of the zymogen molecule
- Cascade of zymogen activation:
  - Caspases to activate apoptosis
  - blood coagulation
  - Digestion of proteins

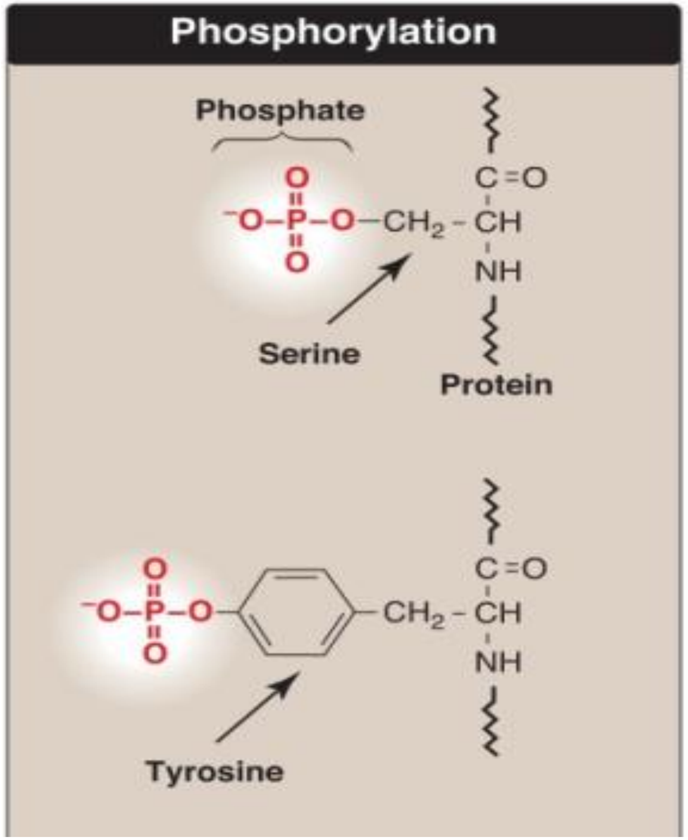


Cleavage of dietary protein in the small intestine by pancreatic proteases (zymogens).

Protein Phosphorylation by a Kinase



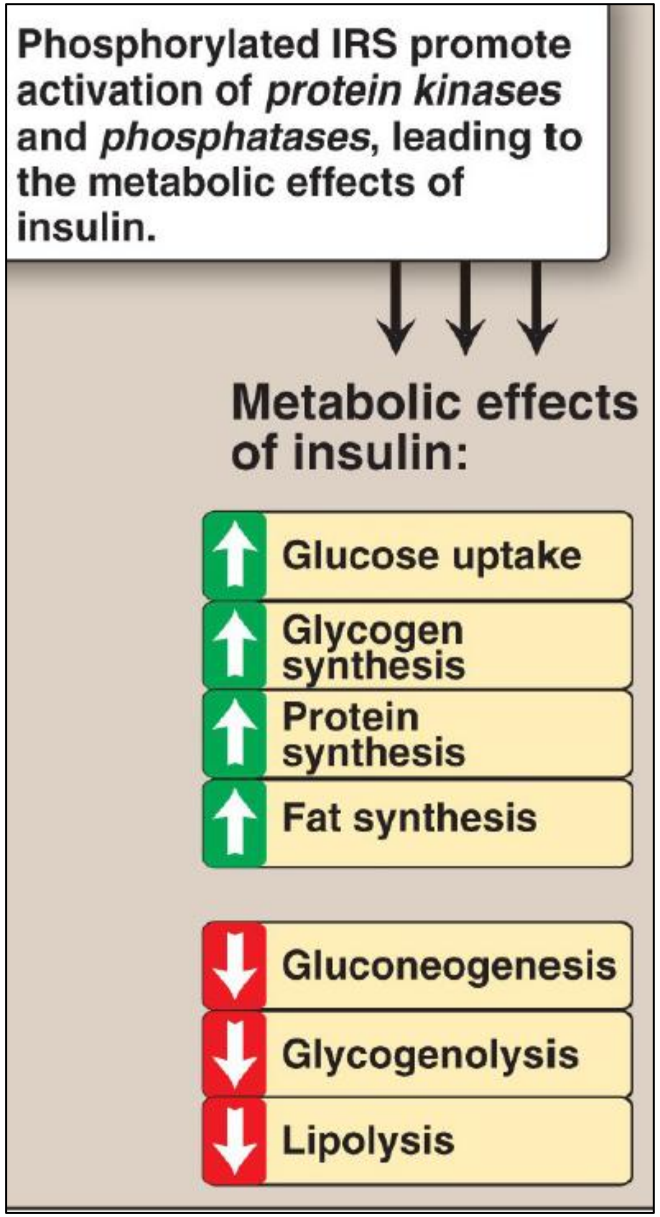
- ✓ Phosphate is transferred from ATP to an amino acid side chain



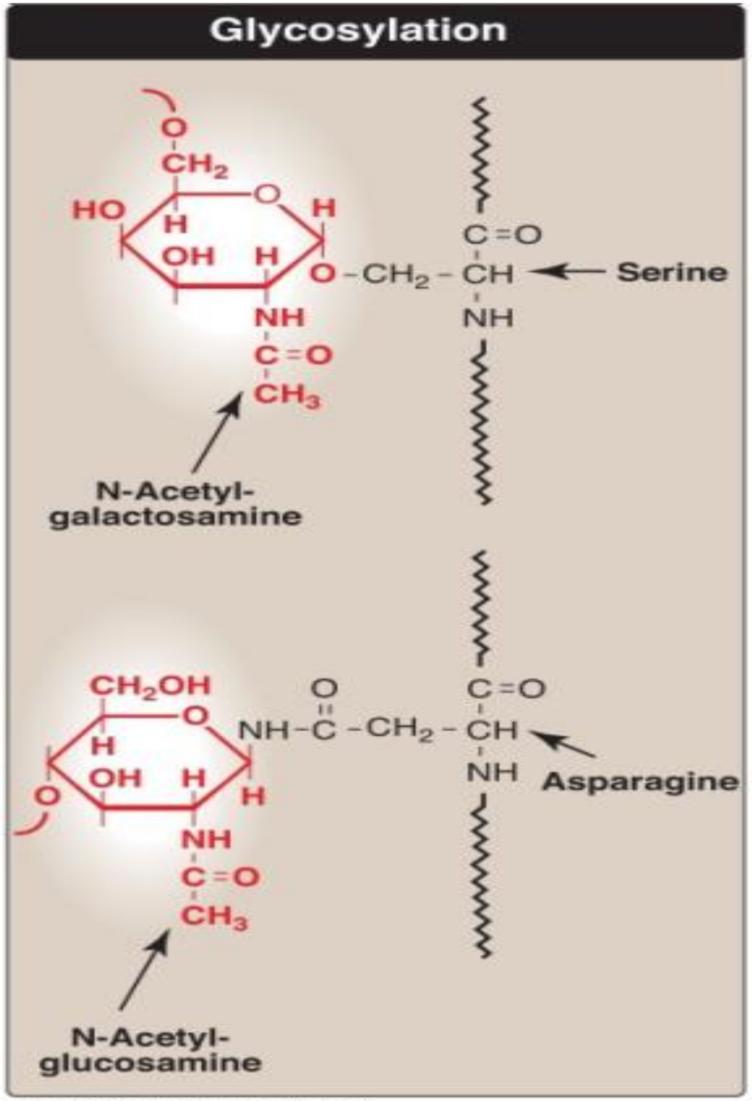
- ✓ Occurs on the OH groups of serine, threonine, or less frequently tyrosine residues.

- ✓ Phosphorylation is the most common post-translational modification in eukaryotes: may be permanent or reversible. Dephosphorylation by phosphatases

- ✓ The Insulin Receptor (IRs) is a tyrosine kinase



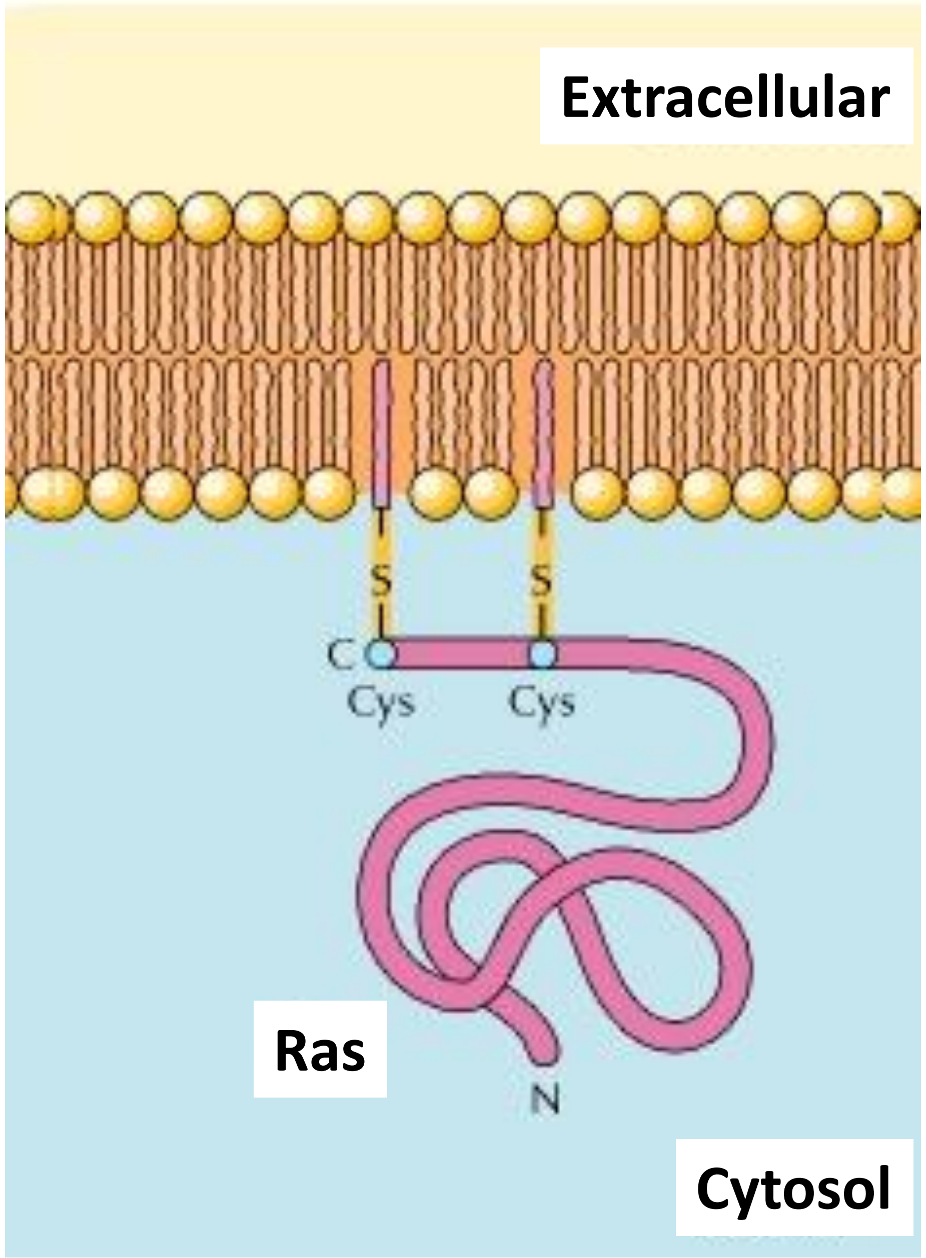
Glycosylation



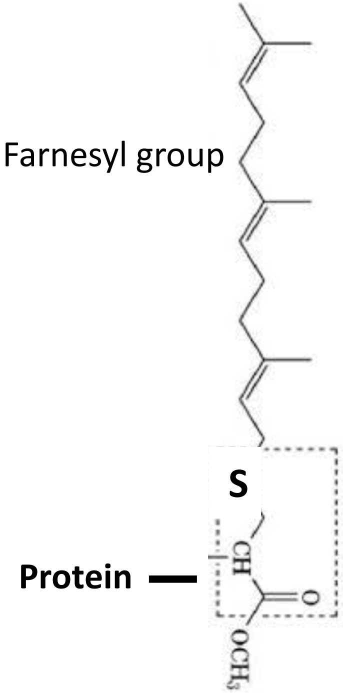
Catalyzed by Glycosyltransferases

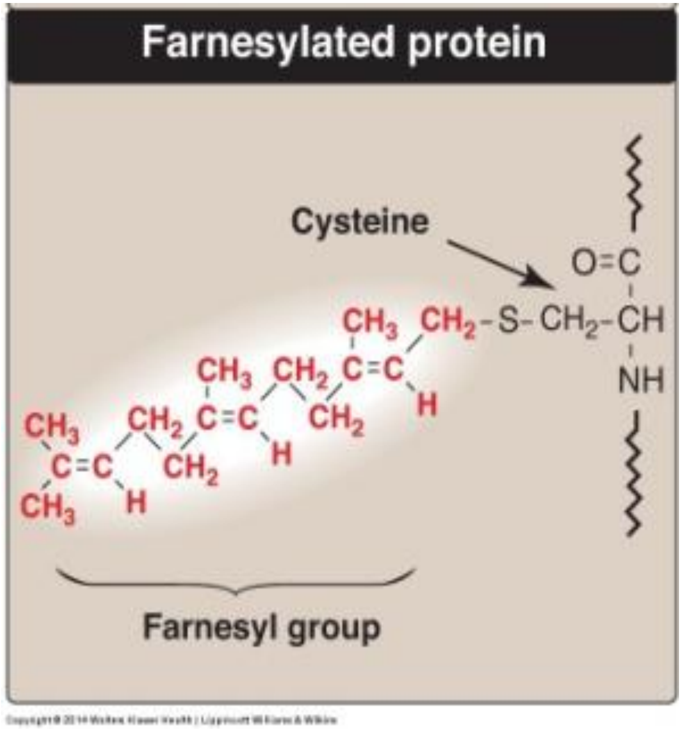
- Alters the properties of proteins, changing their stability, solubility & physical bulk.
- **O – linked**
  - Carbohydrate chain attached to the **OH group of Ser/Thr**
  - Glycan groups always face extracellular side
  - Occurs only after the protein reaches the Golgi Apparatus
- **N – Linked**
  - Carbohydrate chains attached to the amide **nitrogen of Asn residue**.
  - Occurs in the ER & Golgi. In ER, modulates folding of proteins.
- The carbohydrate moieties act as recognition signals:
  - Protein is targeted to either the plasma membrane, to organelle interiors or organelle membranes.
  - Influence cell-cell interactions.
  - Involved in the development of an organism.

Lipid Anchoring



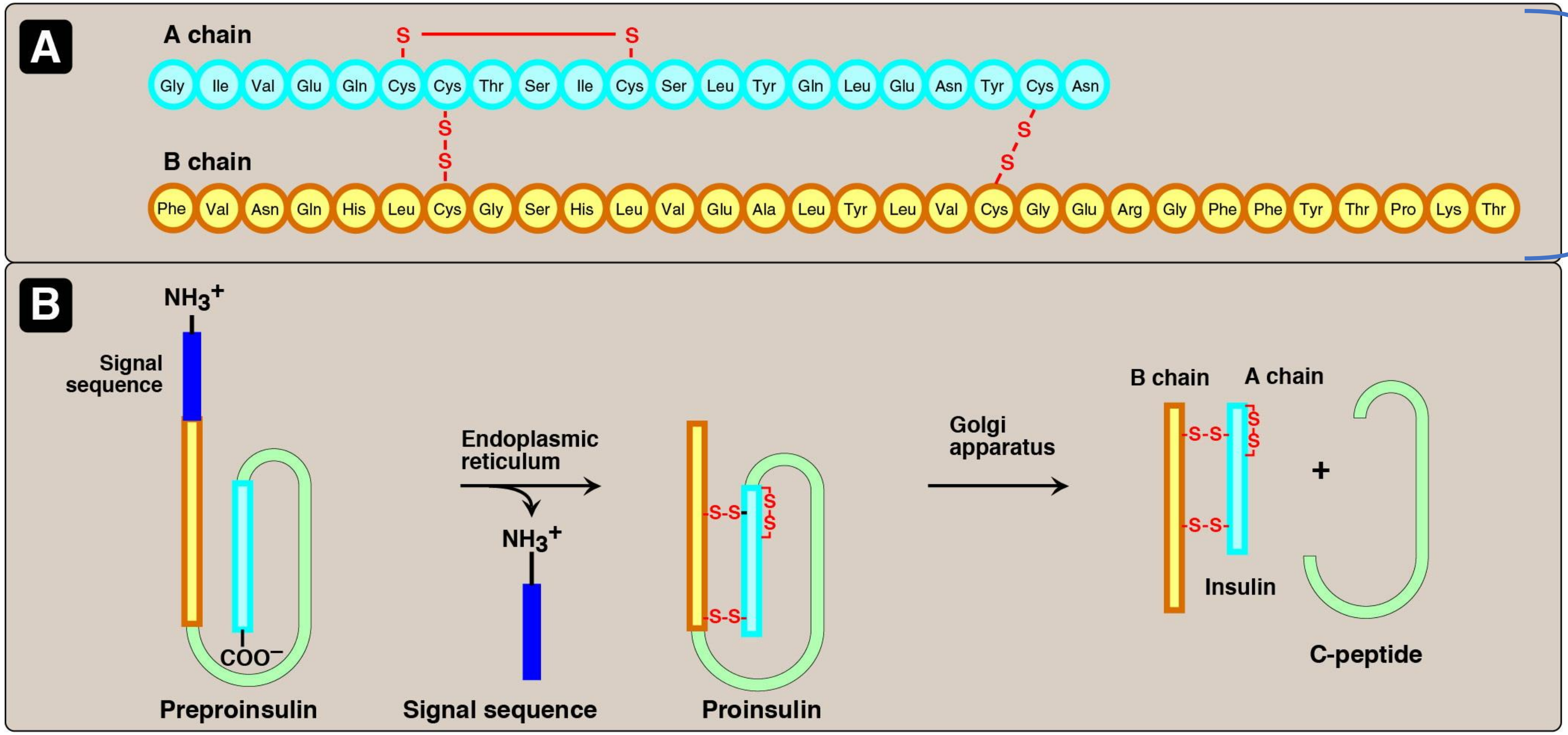
- The cell targets Ras protein to the cytosolic face (inner leaflet) of the plasma membrane by a lipid anchor mechanism - with the aid of farnesyl groups.





- Farnesyl is a 15-carbon isoprenoid group which may be attached to cysteine.

Proteolytic Processing of Insulin

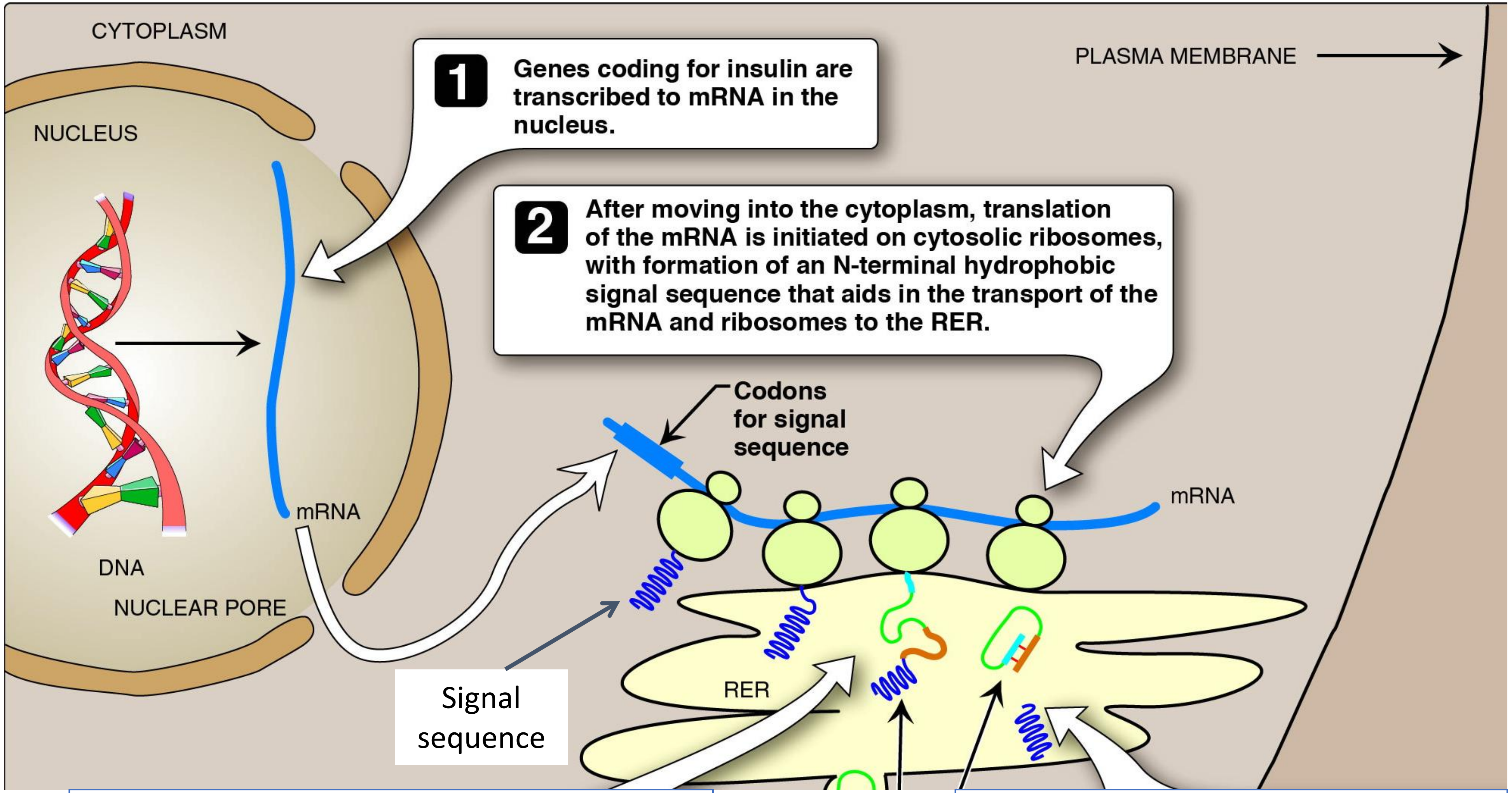


- **Insulin is derived from a single polypeptide**
- **A chain** = 21 amino acids,
- **B chain** = 30 amino acids

- Insulin binds to Insulin Receptor (tyrosine kinase) to signal that blood glucose is high

- C-peptide is essential for proper insulin folding
- **C-peptide is a good indicator of insulin production and secretion** because its (C-peptide) half-life in the plasma is longer than that of insulin.
- Measurement of the C-peptide levels has a clinical value (discussed later in the course)

Proteins destined for secretion are translated into the rER



3. Translation of the polypeptide is directed into the lumen of the rER and forms "**preproinsulin**"

4. The signal sequence is cleaved in the lumen of the rER – forms "**proinsulin**"

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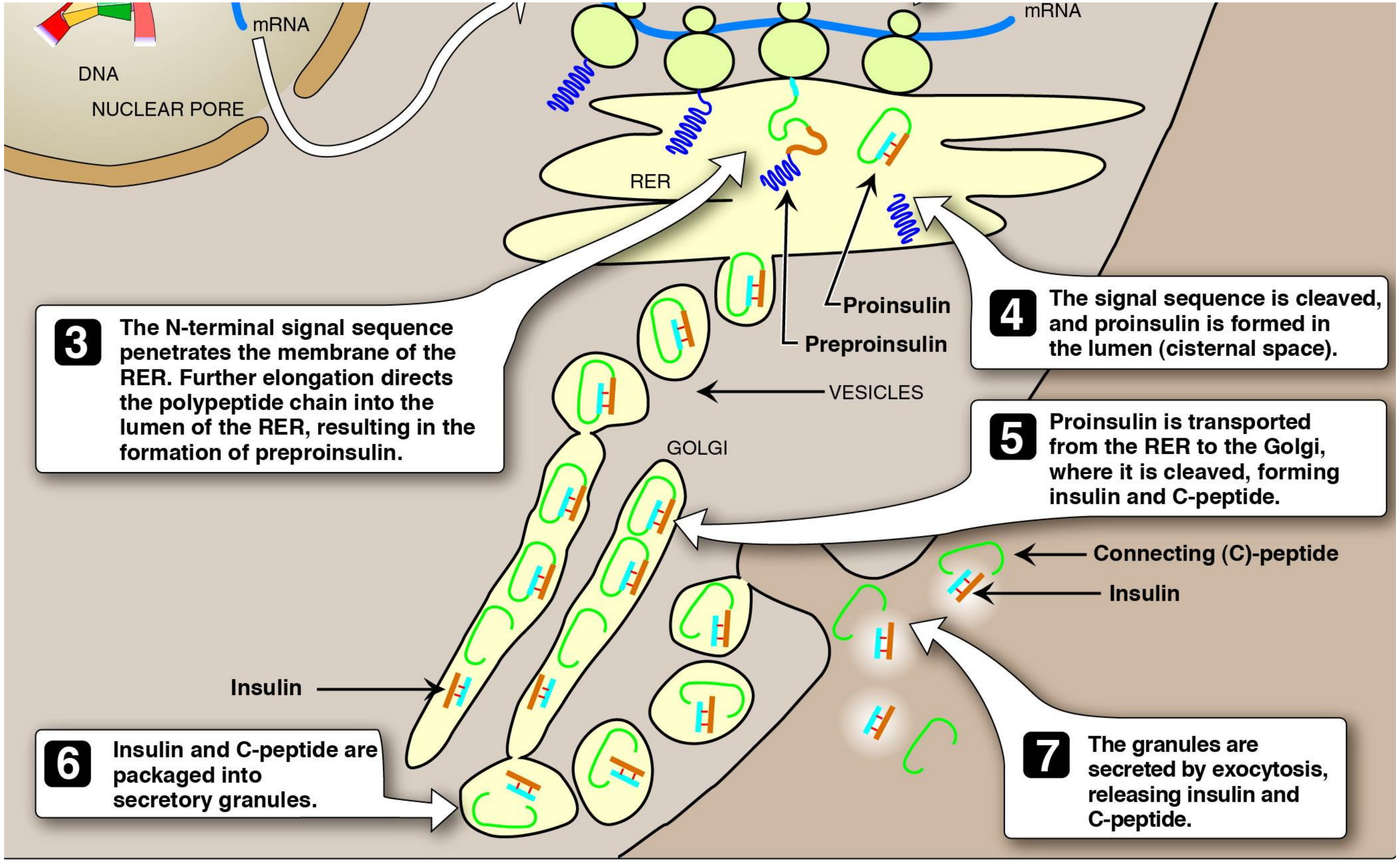
SOM.MK.I.BPM1.1.FTM.3.BCHM.0076

Maturation of insulin occurs in the Golgi



- Insulin is secreted by the β-cell in response to high blood glucose
- Insulin and its C-peptide are packaged together into secretory vesicles to be made ready for secretion
- Insulin and the C-peptide are released into the blood
- There might be (?) a cellular role for the C-peptide

SOM.MK.I.BPM1.1.FTM.3.BCHM.0075 SOM.MK.I.BPM1.1.FTM.3.BCHM.0076

Thank you!

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